

RYAN HARTUNG

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EDUCATION

The University of Notre Dame

Ph.D. in Computer Science & Engineering

- Advisor: Dr. Douglas Thain
- Research focus: Distributed Systems and High Performance Computing

Notre Dame, IN

Expected: May 2029

The University of Notre Dame

Master of Science in Computer Science & Engineering

Notre Dame, IN

August 2025

The University of Texas at Austin

Bachelor of Science in Physics; Computation Option

- Certificate in Elements of Computing

Austin, TX

May 2024

TEACHING

The University of Notre Dame

Graduate Teaching Assistant

Notre Dame, IN

August 2024 - May 2025

- Elements of Computing II
- Principles of Computing

Spring 2025

Fall 2024

PUBLICATIONS

1. Liubov Kurafeeva, **Ryan Hartung**, Benjamin Carter, Alan Subedi, Avhishek Biswas, Michael Fay, Shantenu Jha, Chandra Krintz, Andre Merzky, Douglas Thain, Memet Can Vuran, and Rich Wolski. 2026. Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference. *arXiv preprint arXiv:2605.20532*. <https://arxiv.org/abs/2605.20532>
2. Liubov Kurafeeva, Alan Subedi, **Ryan Hartung**, Michael Fay, Avhishek Biswas, Shantenu Jha, Ozgur Kilic, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, and Rich Wolski. 2025. xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture. In *Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC Workshops '25)*. Association for Computing Machinery, New York, NY, USA, 2317–2327. <https://doi.org/10.1145/3731599.3767589>

TALKS/PANELS

(Nov. 2025) Invited panelist at **XLOOP Workshop at SC '25** [St. Louis, MO] ==> [Watch Here](#)

PRESENTATIONS

GCASR 2026

Attended:

Evanston, IL

May 11, 2026

- Presented the following poster:
 - * Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference

HPDC 2025

Attended From:

Notre Dame, IN

July 20th - July 23rd, 2025

- Presented the following poster:
 - * xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture

RESEARCH EXPERIENCE

The University of Notre Dame	Notre Dame, IN
Graduate Research Assistant	January 2025 - Present
Assisting with the development of xGFabric.	
The University of Texas at Austin	Austin, Texas
Undergraduate Research Assistant	June 2022 - December 2023
- Created an open-source program which leverages various Python packages to analyze star-spots patterns across multiple thousands of stars in a compressed algorithm. Ensured that the algorithm was expandable and efficient for large datasets (over 6,000 star systems) - Compared the contrast ratio of star-spots from two different NASA missions (Kepler and TESS) to establish a basis for exoplanet identification - Implemented parallel computing and multi-core processing to optimize runtime and compilation - Published results to a public GitHub repository - Funded by a NASA research grant	

WORK EXPERIENCE

Hargrove Engineers and Constructors	Katy, TX
Controls & Automations Engineering Intern	May 2024 - August 2024
- Automated the migration of 200,000+ lines of Dow MOD5 code into Honeywell Experion without the loss of functionality - Tracked, analyzed, and documented over 7,000 variables to design and draw comprehensive flow diagrams to visualize variable definitions - Developed multiple algorithms to expedite the migration and variable mapping processes, improving overall efficiency	
Hargrove Engineers and Constructors	Katy, TX
Controls & Automations Engineering Intern	May 2023 - August 2023
- Instantiated existing library modules to complete the upcoming Reactor and Train configuration for Westlake’s PVC plant - Performed safety testing on all module and sequence configurations using internal procedures on simulated ABB hardware - Successfully configured Modbus communication between the ABB and HIMA system controllers - Updated structured text configuration to handle cross-controller communications	

LEADERSHIP AND INVOLVEMENT

• Society of Catholic Scientists	August 2025 - Present
• Science Policy Initiative at Notre Dame	August 2025 - Present
• Catholic Graduate Community	April 2025 - Present
• Technology and Catholicism Club	February 2025 - Present
• Notre Dame Rock Climbing Club Member	August 2024 - Present
• Notre Dame Texas Club Member	August 2024 - Present
• Science Olympiad (Competitor, Mentor, Club President, Proctor, Event-Supervisor, and Lab Coordinator)	August 2011 - Present

TECHNICAL SKILLS

• Proficient Programming Languages: C/C++, HTML, Java, JavaScript, JSON, Julia, Python, Rust, XML
• Machine Learning & AI: Keras, PyTorch, Scikit-learn, TensorFlow
• Distributive Systems: Univa Grid Engine (UGE), HTCondor, SLURM
• Other Tools & Frameworks: ABB, Git, Honeywell, LaTeX, Linux/UNIX, Microsoft Suite, OpenFOAM, ParaView